Silicon Sampling on CES 2025: What We Learned, What Comes Next

One page of findings · the plan for the next round

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1. What we learned so far

Two branches, same test: 10 CES 2025 questions, 15 subgroups (5 regions × 3 ages), nothing from 2025 used except as the answer key. Score: mean **TV**, the share of answers in the wrong place (lower is better). **Justin** prompted an off-the-shelf model (Qwen 3.5 9B) with CES 2021 profiles. **Hubert** fine-tuned Llama 3.3 70B on 50,000 answers from 8 Canadian surveys (2019–2024), then had it answer as Democracy Checkup 2024 respondents described by 7 of their own answers. Temperature 1.0 for both.

Question	CES 2021 unchanged	Fine-tuned Llama 70B	Prompted Qwen 9B	Floor
Spending: affordable housing	**0.08**	0.10	0.13	0.05
Satisfaction with democracy	**0.06**	0.12	0.23	0.05
Medical assistance in dying	**0.09**	0.14	0.23	0.09
Jobs before environment	**0.13**	0.15	0.31	0.10
Spending: defence	0.31	**0.17**	0.24	0.04
Pipelines	**0.16**	0.17	0.31	0.10
Spending: national childcare	**0.09**	0.18	0.31	0.06
Personal finances, past year	**0.13**	0.21	0.21	0.04
Spending: reconciliation	**0.17**	0.24	0.44	0.05
Immigration levels	0.31	**0.30**	0.60	0.04
Mean	**0.151**	0.178	0.302	0.062

- **Fine-tuning works:** 0.325 → 0.178 for the same model before and after; best prompting reaches 0.302.
- **But a 2021 poll reused as is does better (0.151),** on 8 of 10 questions. That is only possible because our 10 questions were asked before.
- **Models flatten subgroups:** right direction, but gaps about a third of their real size (both branches find this, by different methods).
- **Most of the error is the national level** (0.157 of the 0.178).
- **Did not help:** a bigger prompted model, shorter descriptions, picking context by similarity of meaning, 100k instead of 50k examples, stretching model gaps (IRT), and the model's national level combined with CES 2021 gaps (0.165).

2. Next steps

The target. A real poll also misses, by chance, and less so the more people it asks. We simulated real polls of different sizes drawn from CES 2025 and measured their TV the same way. The model is "worth" the poll size that misses by as much.

Mean TV by subgroup	Same error as a real poll of… per subgroup	…in total (15 subgroups)
0.178 (fine-tuned model today)	about 14 people	about 200
0.151 (CES 2021 reused)	about 20	about 300
0.10 (target: useful)	**about 40**	**about 600**
0.08 (target: reliable)	**about 65**	**about 1,000, a standard poll**

One difference: a real poll's error is random and shrinks as it grows; a model's error is a bias and does not shrink with more fake respondents.

A. Change the test: questions never asked before

A client will ask questions that no past poll has asked. Our 10 questions were all in CES 2021, which is why reusing CES 2021 is such a strong rival here: in real use, it would not exist.

- **Add a second set of questions:** CES 2025 items never asked in earlier surveys, and removed from all training data. This is the score that matters for the product.
- **Keep the current set** with CES 2021 as the target to reach (0.151): on repeated questions, a model must at least match a past poll.

B. Make the fine-tuned model know each person better

The models flatten subgroups because they know too little about each person and fall back on the average. Today they see 7 of the person's answers.

1. **Show 15 to 20 answers instead of 7:** a fixed core of themes, plus 5 or so answers picked for their closeness to the question asked. Cost is not a constraint.
2. **Start at 50,000 examples**, then try 100,000 and 150,000 with the best recipe, training each size twice to see how much results vary from one training run to the next.

Temperature stays at 1.0. Raising it makes every group answer more at random, which flattens them further; lowering it pushes everyone toward the most common answer.

C. A new approach: from a dossier to a distribution

> **Today**, the model plays 15,000 fake respondents, one at a time, and we count their answers. **Instead**, the model would act like an analyst: we hand it a **dossier** on the new question and it writes down, in one go, **the percentage of each answer in each of the 15 subgroups**.

What the dossier contains (all from past surveys, nothing about the new question's answers):

- the question's wording and options;
- how Canadians answered **similar past questions**, nationally;
- how **each subgroup differed from the national figure** on those questions (for instance, how far Quebec 18–34 sits from the national figure on each spending item);
- simple summaries of the main attitude dimensions (left–right, trust, economy) by subgroup.

What it returns: a table of 15 subgroups × the answer options, in percentages.

Why it is promising.

- It gives exactly what a client wants, for a question nobody asked yet.
- It uses what we know is stable: subgroups differ in similar ways across related questions.
- One request per question instead of 15,000.
- A published study finds language models **describe** a group's opinion distribution more accurately than they **reproduce** it by role-playing respondents one by one (Meister, Guestrin and Hashimoto, NAACL 2025, *Benchmarking Distributional Alignment of Large Language Models*).

How we would train it. Every question in our 8 surveys becomes one practice exercise: build its dossier without that question, ask the model for the 15 distributions, compare with the real answers. That gives thousands of exercises. Two ways to learn from them:

1. **By example** (as today): the model is shown the right table and learns to write it.
2. **By reward**: the model proposes a table, a small program scores it (the closer to reality, the higher), and the model learns to raise its score. This targets our real goal directly. Azure offers this for some OpenAI models (o4-mini; GPT-5 on invitation), with a scoring program we write ourselves, but not for Llama according to its documentation.

Practice and test questions must cover different topics, or the model would see near-copies of the test questions while practising.

How we would judge it. Against the fine-tuned model of section B, and against a plain statistical version of the same idea with no language model (predict a new question's subgroup gaps from similar questions; Justin's August "item encoder"). If the language model cannot beat that, it adds nothing.

First step, no training needed: build the dossier for our 10 questions as if they were new (leaving them out of it) and ask an off-the-shelf model. About ten requests.

D. Already measured, for later

- **A product option:** one real national figure for the new question (a small national poll) plus the model's subgroup differences gives **0.082**, worth about 60 real respondents per subgroup. The model then only does what it does well: telling how groups differ.
- **For the next note's appendix:** does the model use the survey year? Same person, same question, with "2019" swapped for "2025" in the prompt. If the national answer barely moves, the model answers for an average of 2019–2024.